

D426

Data Management

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A hospital's data team sketches Patient, Physician, and Visit with lines showing how they relate, but no data types, keys, or storage details. The sketch was drawn on a whiteboard during a two-hour workshop. What level of model is this?

- A. Logical
- B. Not a model until it lists data types.
- C. Conceptual
- D. Physical

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

At a training company, each course has many students and each student takes many courses. The company has 14 instructors and 2,300 learners. How should this be built in relational tables?

- A. An Enrollment table with foreign keys to Student and Course, forming a composite primary key.
- B. One table per instructor.
- C. A comma-separated list of course IDs in each Student row.
- D. A CourseID foreign key in the Student table.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A table Enrollment(StudentID, CourseID, Grade) has the primary key (StudentID, CourseID). What relationship does it resolve?

- A. One-to-many from Student to Course.
- B. One-to-one between Student and Course.
- C. Many-to-many between Student and Course.
- D. It shows Grade is its own entity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An online store's Customer table has CustomerID (auto-number), Email, Phone, and LastName. Marketing says emails are reliable and phones rarely change; the store has 80,000 customers. Which column is the best primary key?

- A. CustomerID
- B. LastName
- C. Email + Phone + LastName together.
- D. Email

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Orders.CustomerID references Customers.CustomerID. The DBA runs nightly backups at 2 a.m. A clerk tries to insert an order with CustomerID 9999, which does not exist in Customers. What happens?

- A. The insert succeeds and a blank customer 9999 is created.
- B. The insert is rejected for violating referential integrity.
- C. The insert succeeds and the nightly backup repairs it.
- D. The insert succeeds with CustomerID set to null.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A clinic's Patient table has an Allergies column holding values like "penicillin, latex, peanuts." The clinic moved to a cloud host last year. Which normal form is broken, and what is the fix?

- A. 3NF; split Patient into two tables by name.
- B. 1NF; move allergies to a separate table with one row per patient-allergy pair.
- C. Add Allergy1, Allergy2, and Allergy3 columns.
- D. None, because cloud databases accept text lists.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

OrderLine(OrderID, ProductID, Quantity, ProductName) has the composite key (OrderID, ProductID). The table holds 3 million rows. What design problem does it have?

- A. A transitive dependency, which violates 3NF.
- B. A 1NF violation.
- C. A partial dependency, which violates 2NF.
- D. Too many rows; split the table by year.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A table is in 1NF and its primary key is a single column. Can it violate 2NF?

- A. Yes, always, once it has more than three columns.
- B. Yes, if a non-key column depends on another non-key column.
- C. Only if the table has no foreign key.
- D. No; with a one-column key there is no part of the key to depend on.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Employee(EmpID, Name, DeptID, DeptName) has EmpID as its primary key. The company has offices in five cities. What problem does the table have?

- A. A 1NF violation.
- B. It needs a City column.
- C. A partial dependency (2NF).
- D. A transitive dependency (3NF); move DeptName into a Department table.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

In table T(A, B, C), A is the key, and $A \rightarrow B \rightarrow C$. How should T be decomposed to reach 3NF?

- A. T1(A, B, C) and T2(A, C).
- B. T1(A, B) and T2(B, C), with B as T2's key.
- C. T1(A, C) and T2(B, C).
- D. Keep T and add an index on C.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A developer must add a Birthdate column to an existing Patients table that has 120,000 rows, during a Sunday maintenance window. Which statement does this?

- A.** ALTER TABLE Patients ADD Birthdate DATE;
- B.** DROP TABLE Patients; then CREATE it with the new column.
- C.** UPDATE Patients SET Birthdate = NULL;
- D.** INSERT INTO Patients (Birthdate) VALUES (NULL);

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A developer runs DROP TABLE Temp;. What kind of statement is this, and what does it do?

- A.** DML; it deletes all rows but keeps the table.
- B.** DDL; it removes the structure but keeps the rows for recovery.
- C.** DCL; it removes users' access to the table.
- D.** DDL; it removes the table's structure and all its rows.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A hospital's data team sketches Patient, Physician, and Visit with lines showing how they relate, but no data types, keys, or storage details. ~~The sketch was drawn on a whiteboard during a two-hour workshop.~~ What level of model is this?

Target word: model level **Decides:** Only entities and relationships are shown.

A	Adjacent category	Logical models add attributes and keys.
B	Absolute language	Conceptual models deliberately leave types out.
C	Correct	Entities and relationships without attributes, keys, or types.
D	Inverted operation	Physical is the most detailed level, with types and indexes.

2. Answer A Concept 2

At a training company, each course has many students and each student takes many courses. ~~The company has 14 instructors and 2,300 learners.~~ How should this be built in relational tables?

Target word: implement many-to-many **Decides:** Many on both sides.

A	Correct	A junction table splits many-to-many into two one-to-many links.
B	Camouflage	Instructors are not part of the student-course relationship.
C	Confusion set	Breaks 1NF and cannot be enforced as a foreign key.
D	Dropped element	Allows only one course per student.

3. Answer C Concept 2 · transfer

A table Enrollment(StudentID, CourseID, Grade) has the primary key (StudentID, CourseID). What relationship does it resolve?

Target word: reverse: junction table to relationship **Decides:** Two foreign keys forming the key.

A	Adjacent category	One-to-many needs only a foreign key on the many side.
B	Confusion set	Students and courses both repeat in the table.
C	Correct	Each pair appears once; each side can repeat.
D	Dropped element	Grade is an attribute of the enrollment.

4. Answer A Concept 3

An online store's Customer table has CustomerID (auto-number), Email, Phone, and LastName. ~~Marketing says emails are reliable and phones rarely change; the store has 80,000 customers.~~ Which column is the best primary key?

Target word: best primary key **Decides:** Unique, non-null, stable, minimal.

A	Correct	A surrogate key is unique, never null, and never changes.
B	Confusion set	Many customers share last names.
C	Adjacent category	A superkey, not minimal, and made of changeable values.
D	Camouflage	Customers change emails; the marketing claim does not make it stable.

5. Answer B Concept 4

Orders.CustomerID references Customers.CustomerID. ~~The DBA runs nightly backups at 2 a.m.~~ A clerk tries to insert an order with CustomerID 9999, which does not exist in Customers. What happens?

Target word: result of an orphan insert **Decides:** Foreign key must match an existing parent.

A	Adjacent category	Foreign keys do not create parent rows.
B	Correct	The DBMS blocks child rows that point to no parent.
C	Camouflage	Backups copy data; they do not enforce constraints.
D	Confusion set	The DBMS does not silently swap in null.

6. Answer B Concept 5

A clinic's Patient table has an Allergies column holding values like "penicillin, latex, peanuts." ~~The clinic moved to a cloud host last year.~~ Which normal form is broken, and what is the fix?

Target word: normal form and fix **Decides:** Multiple values in one field.

A	Adjacent category	The problem is non-atomic values, not transitive dependency.
B	Correct	Atomic values restore 1NF and allow querying each allergy.
C	Confusion set	A repeating group, still a 1NF problem.
D	Camouflage	Hosting does not change normalization rules.

7. Answer C Concept 6

OrderLine(OrderID, ProductID, Quantity, ProductName) has the composite key (OrderID, ProductID). ~~The table holds 3 million rows.~~ What design problem does it have?

Target word: identify the dependency **Decides:** ProductName depends only on ProductID.

A	Confusion set	Transitive means non-key → non-key; here it is part-key → non-key.
B	Adjacent category	Every value is atomic.
C	Correct	ProductName depends on part of the key; move it to a Product table.
D	Camouflage	Row count is not a normalization issue.

8. Answer D Concept 6 · transfer

A table is in 1NF and its primary key is a single column. Can it violate 2NF?

Target word: reverse: when 2NF can fail **Decides:** A partial dependency needs a composite key.

A	Absolute language	Column count does not create partial dependencies.
B	Confusion set	That is a 3NF violation.
C	Adjacent category	Foreign keys have nothing to do with 2NF.
D	Correct	2NF only concerns composite keys.

9. Answer D Concept 7

Employee(EmpID, Name, DeptID, DeptName) has EmpID as its primary key. ~~The company has offices in five cities.~~ What problem does the table have?

Target word: identify the dependency **Decides:** EmpID → DeptID → DeptName.

A	Adjacent category	All values are atomic.
B	Camouflage	Office locations do not bear on the dependency.
C	Confusion set	The key has one column, so there is no partial dependency.
D	Correct	DeptName depends on DeptID, a non-key column.

10. Answer B Concept 7 • transfer

In table T(A, B, C), A is the key, and $A \rightarrow B \rightarrow C$. How should T be decomposed to reach 3NF?

Target word: solve for the decomposition **Decides:** B determines C, so B keys the new table.

A	Confusion set	Keeps the redundancy and adds more.
B	Correct	Removes the transitive dependency without losing information.
C	Dropped element	Loses the $A \rightarrow B$ link.
D	Adjacent category	Indexes speed queries; they do not normalize.

11. Answer A Concept 8

A developer must add a Birthdate column to an existing Patients table ~~that has 120,000 rows, during a Sunday maintenance window.~~ Which statement does this?

Target word: structure change **Decides:** Adding a column changes the table's structure.

A	Correct	ALTER is DDL for changing an existing table.
B	Inverted operation	Destroys all 120,000 rows.
C	Confusion set	UPDATE changes values in columns that already exist.
D	Adjacent category	INSERT adds rows, not columns.

12. Answer D Concept 8 • transfer

A developer runs DROP TABLE Temp;. What kind of statement is this, and what does it do?

Target word: classify from the statement **Decides:** DROP is DDL and removes the object.

A	Confusion set	That describes DELETE without WHERE (or TRUNCATE).
B	Dropped element	Rows go with the table.
C	Adjacent category	Access is handled by GRANT and REVOKE.
D	Correct	DROP deletes the object itself.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	11	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	2	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Conceptual, logical, physical models	Conceptual = entities and relationships only; logical = attributes, keys, normalized structure, no product specifics; physical = tables, data types, indexes for a specific DBMS.	Calling a whiteboard entity sketch a logical or physical model.	Whether data types, keys, or storage details are present.
2	Resolving many-to-many	A many-to-many relationship becomes a junction table holding foreign keys to both sides, usually as a composite primary key.	Putting one foreign key in one table, or a list of IDs in a column.	"Each A has many B and each B has many A."
3	Primary keys	A primary key must be unique, never null, stable, and minimal; a surrogate key is often best.	Choosing a natural value that can repeat or change.	Candidate columns like name, email, and an ID.
4	Referential integrity	A foreign key value must match an existing primary key in the parent table (or be null if allowed).	Thinking the DBMS will accept orphans or create the parent.	Inserting a child row that points to a missing parent.
5	First normal form	Every column holds one atomic value; no lists or repeating groups.	Fixing a list with Column1, Column2, Column3.	Comma-separated values in one field.
6	Second normal form	No non-key column may depend on only part of a composite key.	Calling any redundancy a 3NF problem.	A composite key with a column that depends on one part.
7	Third normal form	No non-key column may depend on another non-key column (no transitive dependency).	Confusing transitive with partial dependency.	Key $\rightarrow X \rightarrow Y$ inside one table.
8	DDL vs. DML	DDL (CREATE, ALTER, DROP) changes structure; DML (SELECT, INSERT, UPDATE, DELETE) works with rows.	Using UPDATE or INSERT to add a column.	"Add a column" vs. "change values."

Course scope (public description): Foundations - Data Management Foundations offers an introduction in creating conceptual, logical and physical data models. Students gain skills in creating databases and tables in SQL-enabled database management systems, as well as skills in normalizing databases. No prerequisites are required for this course.